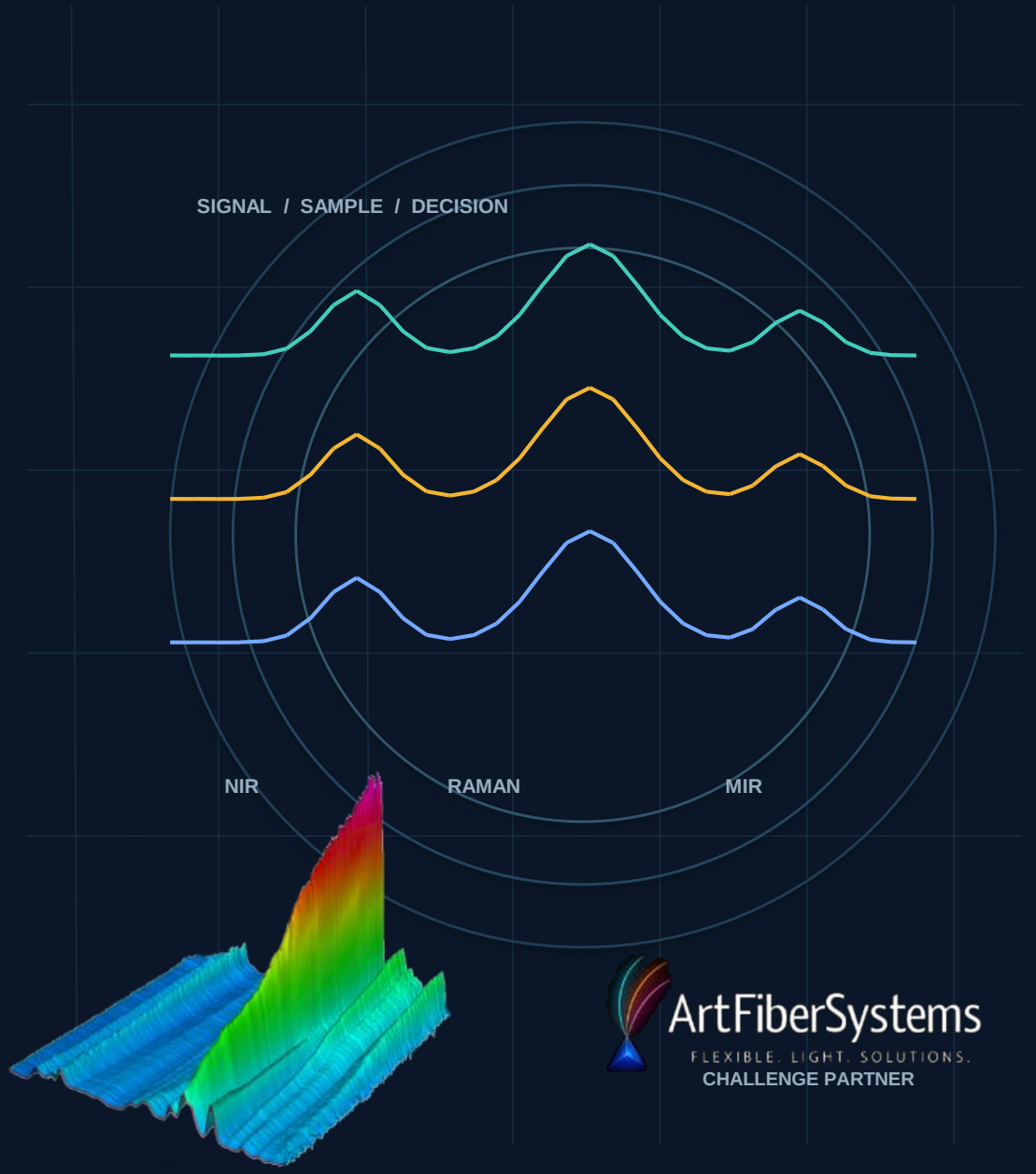
AI FOR INDUSTRIAL SPECTROSCOPY

AFS ASSISTANT
by Team *GUJJU*

Better first decisions.
Grounded in real evidence.

Turns a customer question into a first plan
an AFS expert can review.

AFS SOURCES | SOURCE-LINKED ANSWERS | EXPERT SIGNS OFF



ArtFiberSystems
FLEXIBLE. LIGHT. SOLUTIONS.
CHALLENGE PARTNER

The question is simple. The choice isn’t.

Customers know what they need to measure.
Choosing how to measure it is the hard part.

CUSTOMER NEED

“Measure water in our process line.”

A useful goal — but not yet a measurement plan.



WHAT THE EXPERT MUST DECIDE

- NIR
- RAMAN
- MIR / FTIR
- UV–VIS
- FLUORESCENCE

Which method? Which fibre and probe? What should we test first?

A wrong early choice can waste time and money — or create safety risks.

THE SOLUTION

One enquiry. Six useful outputs.

A clear first plan that an expert can review and improve.

01 What needs measuring

What needs to be measured, and in what conditions?

02 What we still need to know

What details are missing before we choose?

03 Options to compare

Which measurement methods are worth trying?

04 Suggested setup

What equipment could work in this process?

05 Risks and unknowns

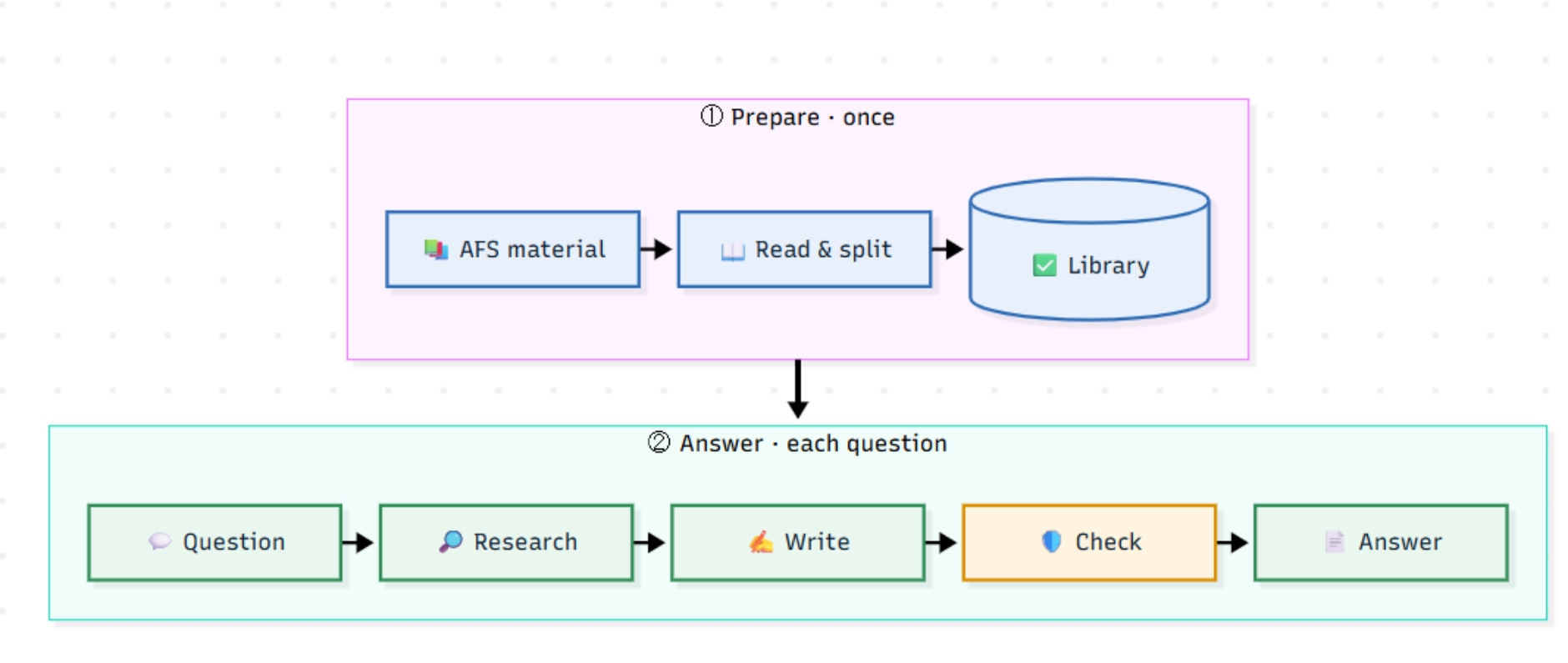
What could go wrong, and what do we still not know?

06 First test and sources

A first test plan, with links to the supporting sources.

Four steps from customer question to reviewed answer.

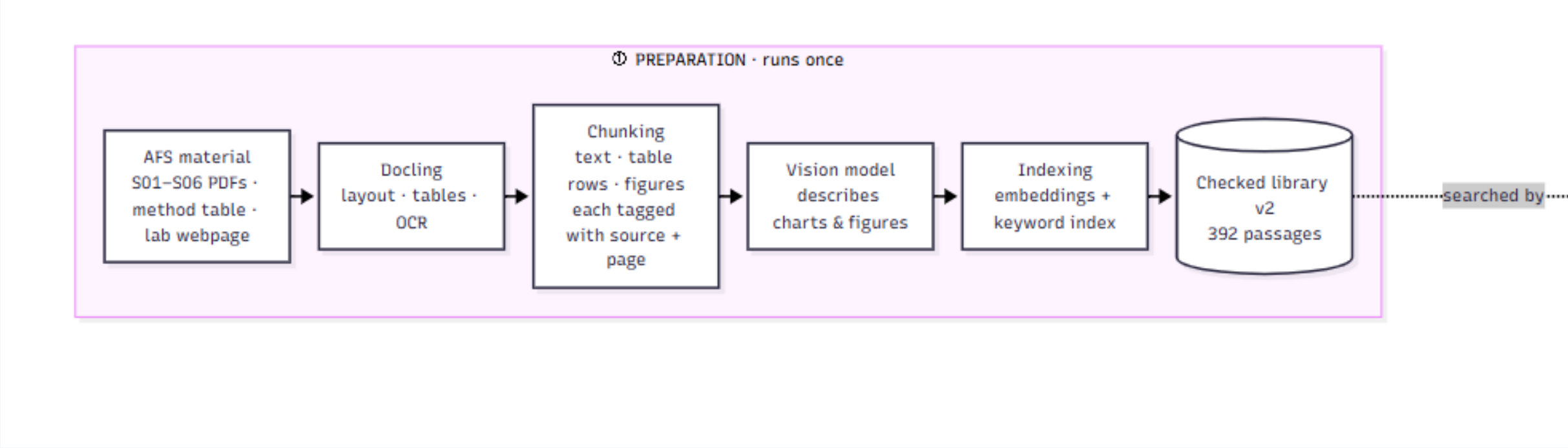
Uses AFS documents as evidence. An expert makes the final decision.



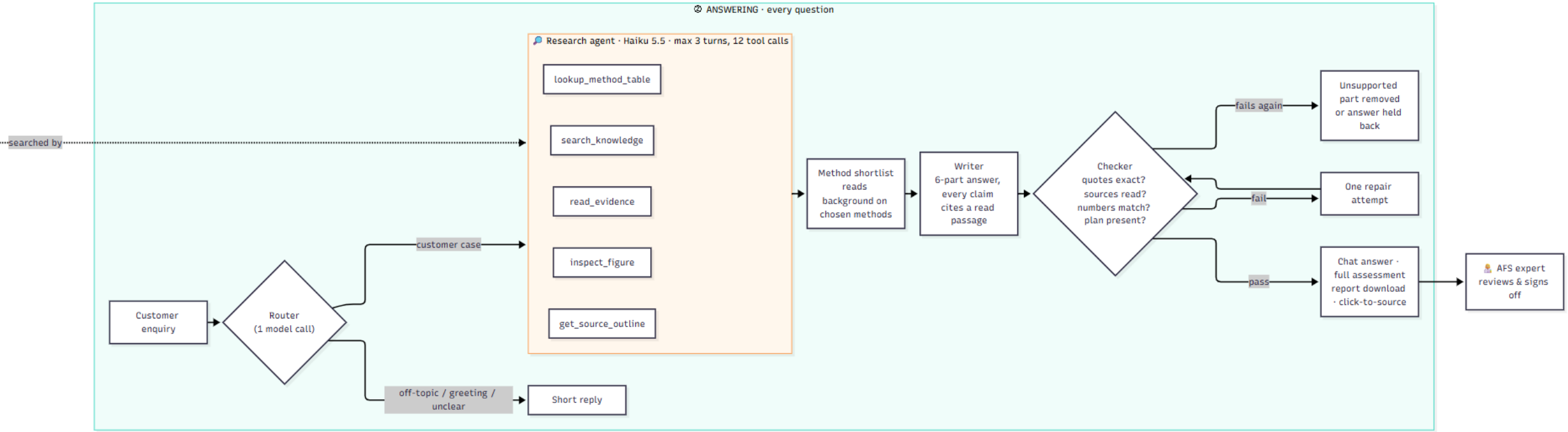
The assistant prepares a first draft. An AFS expert reviews and approves it.

How we prepare AFS documents for search

We organise reports, tables and figures into searchable pages with source links.



What happens when a customer asks a question



TRACEABLE ANSWER

You can check where an answer comes from.

The answer links to the original AFS pages, so an expert can check the evidence for themselves.

AFS SOURCE PAGE

[3] S01 · pp. 15–16

Multispectral Laboratory

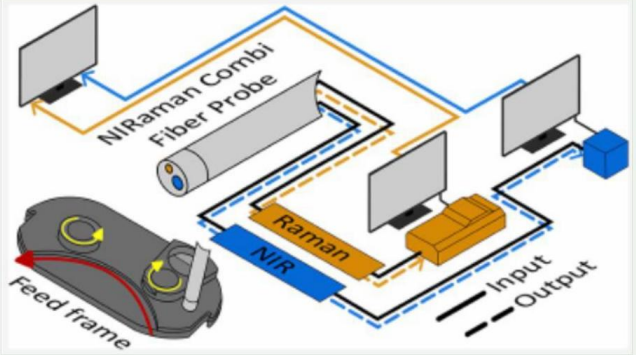
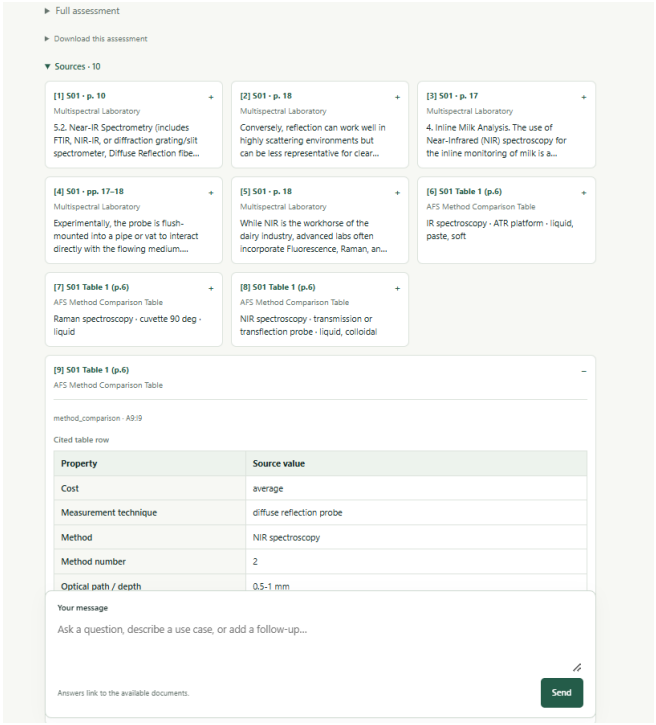


Fig.
6.
A multispectral NIRaman fiber probe has industrial grade design that combines NIR reflection and Raman optical techniques in a compact shaft that fits in a tablet press feed frame.

ASSISTANT RESPONSE



FOLLOW THE SOURCE

01 Source
S01 — Multispectral Laboratory

02 Pages
15–16

03 What it shows
A probe that combines NIR and Raman measurements.

04 Expert check
Open the cited pages to check the claim in context.

It can also flag a bad assumption or out of scope question — for example, a standard silica cable is not suitable for mid-infrared use.

MODEL COMPARISON

Which model finds and cites the right pages?

Average across D01–D08 · compared with the organiser’s list of useful source pages

MODEL	PAGES CHECKED	PAGES CITED	TIME PER QUESTION
Qwen 3.6 · 35B-A3B qwen3.6:35b-a3b	**88.9%** 32 / 36 pages	**72.2%** 26 / 36 pages	**90–120 s**
Claude Haiku 5.5 claude-haiku-5-5	**94%** 34 / 36 pages	**87%** about 31 / 36 pages on average	**60–90 s**

HOW TO READ THIS

Pages checked = suggested pages that matched the source list.

Pages cited = pages included in the answer. 36 expected source pages across D01–D08.

Search model: qwen3-embedding:0.6b. It finds relevant passages; the language model writes the answer.

LIMITS AND EXPERT REVIEW

Useful first guidance — not the final decision.

The assistant helps plan the next step, but an expert must verify the science and the results.

A citation does not prove a claim

The system checks whether sources are cited, but it cannot independently confirm that every scientific statement is correct. An AFS expert must review the source and its context.

CHECK THE SOURCE AND SCIENTIFIC CONTEXT

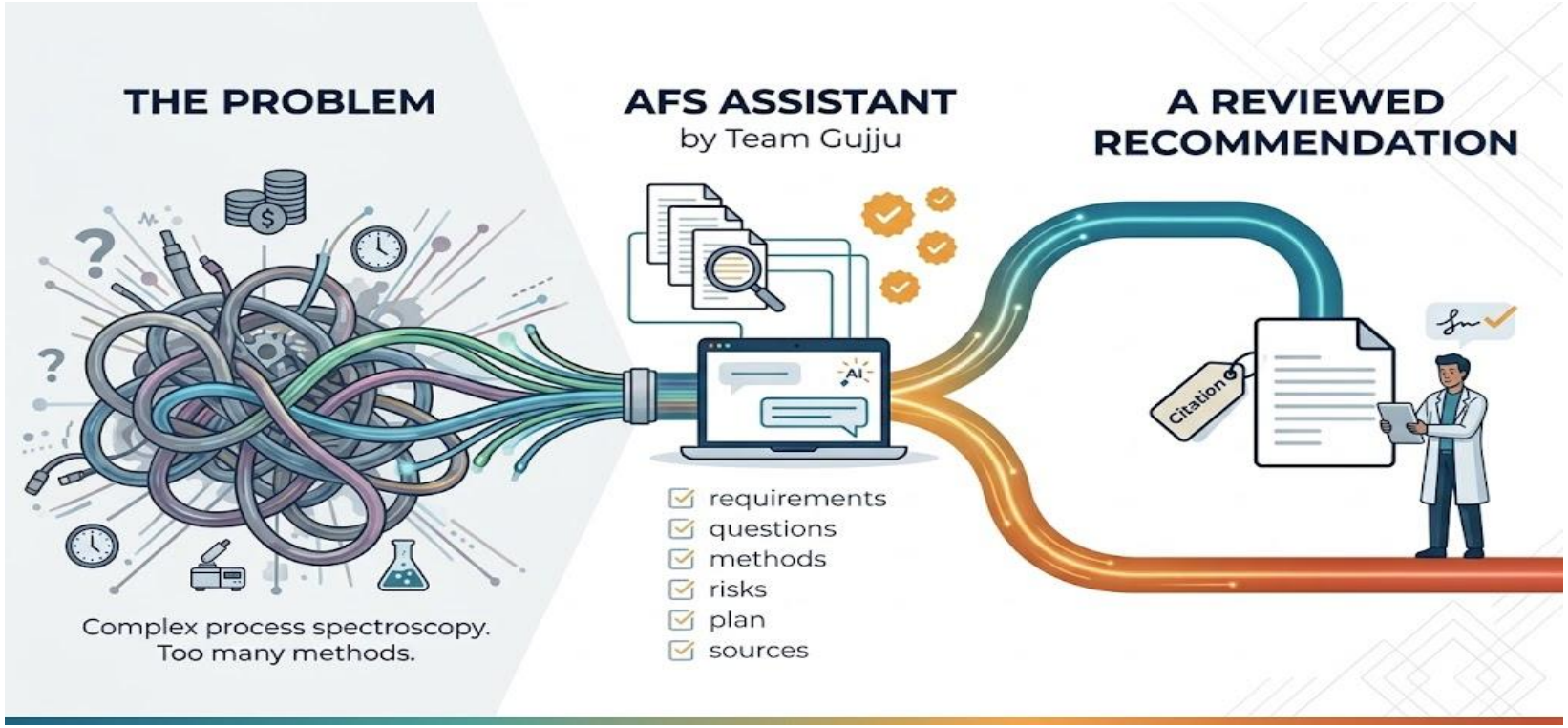
The recommendation still needs testing

The assistant can suggest what to test and how to compare options, but it cannot promise that a setup will meet the required accuracy. Lab tests and an AFS expert make the final call.

CONFIRM WITH LAB TESTS AND AFS EXPERTS

Use the assessment to plan the next experiment, then confirm the result with source evidence and lab tests.

That’s how we solved it in the first iteration!



Thank You!

[Github Repo]